

Safety

Claude edition — the boundary comes first

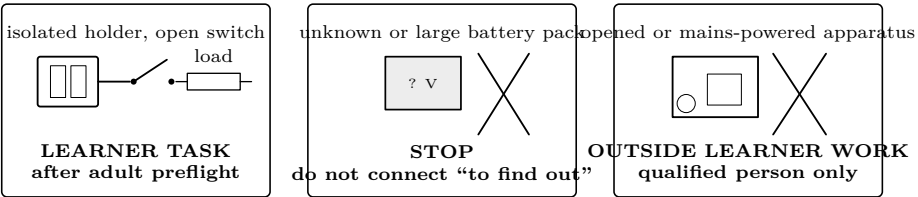
Bound the task, rehearse it cold, then collect evidence

LEARNER LIMIT

Learner investigations use only isolated battery sources at 12 VDC or less. Outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, and exposed high voltage are not learner work. An adult facilitator controls the bounded battery activity; adult status does not make anyone electrically qualified.

Three roles—never collapse them	
Learner	Predicts, assembles only the approved battery circuit while deenergized, observes, records, and obeys every stop call.
Adult facilitator	Selects materials, verifies the boundary and meter setup, owns the switch and timing decision, and ends the trial at uncertainty.
Qualified person	Performs work whose voltage, equipment, environment, or rules require electrical qualification. Being the learner’s parent, teacher, or supervising adult does not confer this role.

Sort the task before touching parts



Boundary questions

1. The label says 9 V, but the pack is large and its current capability is unknown. Which part of the boundary rejects it?
2. An appliance is unplugged. Does that remove stored energy, damaged insulation, or the need for qualification?
3. An adult offers to make the connection. Does changing the hands performing the work change the work’s category?

Preflight: point, name, verify

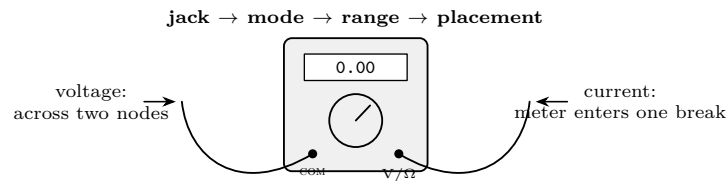
Gate	Learner points and says	Adult verifies
Source	Type, cell count, stated voltage	Isolated holder; correct intact cells; no swelling, leaks, dents, corrosion, heat, or mixed chemistries
Workspace	Switch, disconnect, parts tray	Dry, bright, clear of drinks, jewelry, loose metal, and bystanders; eye protection in place
Conductors	Every intended bare end	Insulation sound; strands contained; no possible bridge across source terminals
Load	Part name and expected behavior	Correct value/rating; LED has its series resistor; coil or resistor shows no prior heat damage
Control	Who owns the switch; trial duration	Switch begins open; disconnect reachable; timer and cool-down interval ready
Evidence	Predicted sign, direction, or range	Prediction is specific enough to distinguish expected, null, and surprising results

DECISION

If a label, rating, connection, meter setting, or expected result is uncertain, the source remains disconnected. “Probably” is a failed preflight.

Checkpoint. One copper strand nearly reaches the other terminal. It has not touched. Explain why this is already a failed conductor gate.

Meter rehearsal—battery holder empty



1. Inspect the case, lead insulation, and probe guards. Remove a damaged meter or lead from service; tape is not a bench repair.
2. Put black in COM. Put red in the jack named by the lesson. Select the quantity and a range that can contain the predicted value.
3. Rehearse voltage by touching the two named nodes with the empty holder. Rehearse current by opening the named path and inserting the meter there.
4. Never place a current-configured meter across a source. Return the red lead from a current jack when the current measurement ends.
5. Adult introduces one cold fault: wrong jack, wrong mode, open connection, or reversed LED. Learner finds it by inspection, never by trial power.

Intermediate meter questions

The display remains zero. Could that mean a true zero, open switch, loose connection, wrong range, or wrong placement? What must happen before moving a lead? **Open the switch and disconnect.**

The expected current is 18 mA. The probes are laid across the holder. Why must the adult stop the trial even though the source is within the voltage limit?

One controlled energizing cycle

1. Adult completes the preflight and calls “ready.” Learner’s hands clear the circuit. The switch owner calls “on” and starts the stated interval.
2. Observe only the planned quantity. Never reposition, squeeze, substitute, or add a part while energized.
3. Switch open; source disconnected; adult checks for unexpected heating. Only then may a lead, probe, load, core, or magnet move.
4. Record the result with units and trial time. A null or surprising result begins a deenergized diagnosis; it never authorizes more voltage.

STOP MEANS THE TRIAL IS OVER

Stop for unexpected warmth, a hot cell or lead, swelling, leakage, sharp odor, smoke, discoloration, buzzing, sparking, pain or tingling, damaged insulation, a dropped metal object, unstable readings, or behavior outside the prediction. Open the switch if safe, disconnect without touching the suspect part, isolate the setup, and tell the adult. Do not repeat “just once.”

Fault decisions

Observation	Immediate action	Cold diagnosis
No response	Open, disconnect, preserve layout	Compare each node with the drawing; check continuity and polarity; do not raise source voltage
Meter overload or jumping digits	Disconnect before touching probes	Recheck jack, mode, range, placement, and predicted magnitude
Part warms more each trial	Stop and leave it untouched	Check value, wiring, current, duration, and cool-down; replace damaged parts
Cell heats, leaks, smells, dents, or swells	Adult clears and isolates the area	Follow cell-maker and local disposal guidance; never reuse or recharge it
Metal bridges source terminals	Disconnect if safe; never grab hot metal	Replace damaged holder, switch, cells, or leads; rebuild from the start
Small magnet or clip is missing	Stop and count all pieces	Find it before anyone leaves; store magnets inaccessible to children

Falsification questions

1. What evidence would show that an apparent “bad resistor” is actually an open switch or lead?
2. Three timed coil trials lift fewer clips each time. Name a thermal change that could matter even though turns and source stayed the same.
3. Why does an unexpected result count against the setup, not against the safety boundary?

Cold verification drill

Use an approved lesson circuit with the holder empty. Learner speaks and points; adult evaluates without prompting.

1. State source type and voltage; identify switch, load, and disconnect.
2. Trace the intended current path and the most plausible accidental short.
3. Demonstrate both meter placements and restore the red lead to V/ Ω afterward.
4. Name five stop observations and show where hands go when “stop” is called.
5. Find an adult-created harmless fault by working through the gates in order. Do not insert cells to diagnose it.

VERIFICATION—EVERY BOX BEFORE CELLS ENTER

☐ Isolated source ☐ ≤ 12 V DC ☐ Correct parts ☐ Intact cells/leads ☐ Switch open ☐ Shorts removed ☐ Meter verified ☐ Prediction written ☐ Time stated ☐ Stop signs named

Final evidence record

Boundary decision

Fault found cold

Meter rehearsal

Stop response

Adult verification

The governing habit

Disconnect before changing anything. Uncertainty is itself a stop condition. Counting magnets before and after, returning meter leads to their normal jacks, and writing down null results are part of the investigation, not cleanup.

Safety basis: OSHA 29 CFR 1910.333 and OSHA Publication 3075; NIST SI references; U.S. Department of Energy electrical safety handbooks. Full citations and scope notes are recorded in the provider-neutral electricity research library.